

AISHA EVERING

Software Engineer | Computational Biology & Bioinformatics

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PROFESSIONAL SUMMARY

Software engineer with 20+ years of experience building production applications and graduate training in computational life sciences. Experienced in applying Python and R to biological data analysis, including transcriptomics, phylogenetics, genomics, statistical analysis, and data processing. Combines full-stack software engineering expertise with hands-on computational biology research to develop reliable tools, workflows, and analysis for complex biological data.

PROJECTS / PORTFOLIO

BIOTOOLS - BIOINFORMATICS COMMAND BUILDER

Designed and developed an interactive web application that translates user-selected alignment filtering criteria into reproducible SAMtools view commands, including SAM flag handling, quality thresholds, presets, and command decoding. Built with React and TypeScript and deployed as a production web application at bio-tools.dev.

DMC1 EVOLUTIONARY ANALYSIS IN METAMONADA

Conducted an end-to-end phylogenetic analysis of the meiotic recombination protein Dmc1 across eukaryotes, including ortholog identification, sequence curation and pruning, multiple-sequence alignment, and maximum-likelihood tree inference and visualization using EukPhylo, Taxus, IQ-TREE, FigTree, and HPC/Slurm workflows. Interpreted gene duplication and loss patterns across metamonad lineages and contributed phylogenetic analysis, scientific writing, and figure development to a collaborative research manuscript.

TRANSCRIPTOMICS AND DIFFERENTIAL GENE EXPRESSION ANALYSIS

Analyzed RNA-seq count data in R using DESeq2, including sample metadata integration, experimental design, dispersion estimation, differential expression testing, and interpretation of log2 fold changes and statistical significance. Applied statistical modeling and data visualization techniques to evaluate gene expression differences across experimental conditions.

ANCESTRY-RELATED DISPARITIES IN POLYGENIC RISK SCORE PERFORMANCE FOR TYPE 2 DIABETES

Investigated ancestry-related disparities in polygenic risk score performance for type 2 diabetes through a structured analysis of published studies, comparing predictive performance, population representation, and cross-ancestry transferability. Evaluated how training data composition and population differences affect model generalizability and identified persistent limitations in the equitable application of PRS across diverse populations.

BIOLOGICAL DATA STANDARDIZATION AND EVIDENCE MAPPING

Designed and implemented Python-based data processing pipelines to clean, normalize, and standardize heterogeneous research method data, improving data quality and enabling downstream analysis and evidence mapping.

SKILLS

BIOINFORMATICS & COMPUTATIONAL BIOLOGY

RNA-Seq, Differential Gene Expression, DESeq2, Phylogenetic Analysis, Multiple Sequence Alignment, Sequence Curation, IQ-Tree, FigTree, EukPhylo, Taxus, SAMtools

PROGRAMMING & DATA ANALYSIS

Python, R, Pandas, NumPy, SQL, Scikit-learn, ggplot2, Matplotlib, Plotly, Tableau, Jupyter Notebook

SOFTWARE ENGINEERING

TypeScript, JavaScript, React, C#, Java, Dart, Git, REST APIs, Full-Stack Development

COMPUTING & MACHINE LEARNING

HPC, Slurm, Linux, Google Cloud, TensorFlow, Pytorch

EXPERIENCE

SOFTWARE ENGINEER, WORKIVA

Scottsdale, AZ

Jan 2019 - Jan 2023

- Developed and enhanced features for a large-scale, customer-facing SaaS platform, translating product requirements into scalable, production-ready software.
- Built and maintained full-stack application functionality and data-processing workflows using Dart, JavaScript, and supporting technologies across front-end and backend systems.
- Designed solutions for transforming, integrating, and manipulating structured data to support application functionality, performance, and data-driven workflows.
- Collaborated with cross-functional Agile teams to design, test, troubleshoot, and deliver reliable software in a complex production environment.

SOFTWARE DEVELOPER, US AIRWAYS/AMERICAN AIRLINES

Phoenix, AZ

Jan 2010 - Jan 2019

- Developed and maintained large-scale, customer-facing web applications for [USAirways.com](https://www.usairways.com) and [AA.com](https://www.aa.com), supporting high-volume customer traffic using C#, [ASP.NET](https://docs.microsoft.com/en-us/aspnet/), Java, and related front-end and backend systems.
- Designed and implemented application features, backend services, and system integrations within complex enterprise environments, with an emphasis on performance, reliability, and maintainability.
- Contributed to large-scale system integration and migration initiatives following the US Airways-American Airlines merger, modernizing applications and improving data flow across platforms.
- Collaborated with geographically distributed, cross-functional teams throughout the software development lifecycle to design, troubleshoot, test, and deploy production systems.

EDUCATION

ARIZONA STATE UNIVERSITY

Master of Science, Computational Life Sciences - **Expected Oct 20226**

UNIVERSITY OF ADVANCING TECHNOLOGY

Bachelor of Science, Software Engineering - May 2001

Associate of Science, Application Development - May 2000

TEXAS STATE TECHNICAL COLLEGE

Associate of Science, Laser Electro-Optics - May 1997